Extra Dessert #17: Test Equipment

Digital oscilloscopes, spectrum analyzers, antenna analyzers, frequency counters, probe compensation, SWR measurement.

Group E4A · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Digital Oscilloscope

Question	Answer	Q
What limits the highest frequency a digital scope can display accurately?	**Sampling rate of the analog-to-digital converter** (E4A01). Nyquist applies: need at least 2x the signal frequency.	E4A01
Effect of aliasing on a digital scope display	**A false, jittery low-frequency version of the waveform is displayed** (E4A06). Same aliasing pattern as in SDR.	E4A06
How to compensate an oscilloscope probe	**Display a square wave and adjust the probe until the horizontal portions are as nearly flat as possible** (E4A04). Built-in cal output is for this purpose.	E4A04
Good practice with a scope probe	**Minimize the length of the probe's ground connection** (E4A09). Long ground leads add inductance, ring the signal.	E4A09
Best trigger mode for measuring linear-supply ripple	**Line** trigger (E4A10). Synchronizes the sweep to the AC mains, so 60/120 Hz ripple stands still.	E4A10

Spectrum Analyzer

Question	Answer	Q
Spectrum analyzer's display axes	**Vertical = signal amplitude, horizontal = frequency** (E4A02). Opposite of a scope.	E4A02
Tool to display spurious signals and IMD products from an SSB transmitter	**Spectrum analyzer** (E4A03). The classic two-tone IMD test instrument.	E4A03

Frequency Counter

Purpose of a prescaler with a frequency counter: Reduce the signal frequency to within the counter's operating range (E4A05). A divide-by-N stage in front of the counter so it can measure microwave inputs.

Antenna Analyzer

Question	Answer	Q
Advantage of an antenna analyzer over an SWR bridge	**Antenna analyzers compute SWR and impedance automatically** (E4A07). Numbers, not a needle that you read against a TX you must key.	E4A07
What can be measured with an antenna analyzer?	Velocity factor + cable length + resonant frequency of a tuned circuit – **all of the above** (E4A11).	E4A11
Tools that can measure SWR	Directional wattmeter + vector network analyzer + antenna analyzer – **all of the above** (E4A08). Three different ways to land on the same SWR number.	E4A08

Dessert 17 Question Bank (E4A)

E4A01. Which of the following limits the highest frequency signal that can be accurately displayed on a digital oscilloscope?
A. Sampling rate of the analog-to-digital converter.
B. Analog-to-digital converter reference frequency.
C. Q of the circuit.
D. All these choices are correct.

E4A02. Which of the following parameters does a spectrum analyzer display on the vertical and horizontal axes?
A. Signal amplitude and time.
B. Signal amplitude and frequency.
C. SWR and frequency.
D. SWR and time.

E4A03. Which of the following test instruments is used to display spurious signals and/or intermodulation distortion products generated by an SSB transmitter?
A. Differential resolver.
B. Spectrum analyzer.
C. Logic analyzer.
D. Network analyzer.

E4A04. How is compensation of an oscilloscope probe performed?
A. A square wave is displayed, and the probe is adjusted until the horizontal portions of the displayed wave are as nearly flat as possible.
B. A high frequency sine wave is displayed, and the probe is adjusted for maximum amplitude.
C. A frequency standard is displayed, and the probe is adjusted until the deflection time is accurate.
D. A DC voltage standard is displayed, and the probe is adjusted until the displayed voltage is accurate.

E4A05. What is the purpose of using a prescaler with a frequency counter?
A. Amplify low-level signals for more accurate counting.
B. Multiply a higher frequency signal so a low-frequency counter can display the operating frequency.
C. Prevent oscillation in a low-frequency counter circuit.
D. Reduce the signal frequency to within the counter's operating range.

E4A06. What is the effect of aliasing on a digital oscilloscope when displaying a waveform?
A. A false, jittery low-frequency version of the waveform is displayed.
B. The waveform DC offset will be inaccurate.
C. Calibration of the vertical scale is no longer valid.
D. Excessive blanking occurs, which prevents display of the waveform.

E4A07. Which of the following is an advantage of using an antenna analyzer compared to an SWR bridge?
A. Antenna analyzers automatically tune your antenna for resonance.
B. Antenna analyzers compute SWR and impedance automatically.
C. Antenna analyzers display a time-varying representation of the modulation envelope.
D. All these choices are correct.

E4A08. Which of the following is used to measure SWR?
A. Directional wattmeter.
B. Vector network analyzer.
C. Antenna analyzer.
D. All these choices are correct.

E4A09. Which of the following is good practice when using an oscilloscope probe?
A. Minimize the length of the probe's ground connection.
B. Never use a high-impedance probe to measure a low-impedance circuit.
C. Never use a DC-coupled probe to measure an AC circuit.
D. All these choices are correct.

E4A10. Which trigger mode is most effective when using an oscilloscope to measure a linear power supply's output ripple?
A. Single-shot.
B. Edge.
C. Level.
D. Line.

E4A11. Which of the following can be measured with an antenna analyzer?
A. Velocity factor.
B. Cable length.
C. Resonant frequency of a tuned circuit.
D. All these choices are correct.

Answer Key

E4A01: A	E4A02: B	E4A03: B	E4A04: A
E4A05: D	E4A06: A	E4A07: B	E4A08: D
E4A09: A	E4A10: D	E4A11: D	